VAAL UNIVERSITY OF TECHNOLOGY



AI-BASED LEARNER STREAM RECOMMENDATION SYSTEM

Project Planning and Implementation Documentation

FACULTY: Applied and Computer Sciences

DEPARTMENT: Computer Sciences

SUBJECT: Business Analysis 3.2

SUBJECT CODE: AIBUY3A

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GitHub Repository:

https://github.com/Comfort-RSA/BA-PROJECT-2026

1. PROJECT OVERVIEW

The AI-Based Learner Stream Recommendation System is a web-based application developed to help Grade 9 learners explore suitable academic streams when preparing to enter Grade 10. Choosing a school stream is an important decision because it can influence the subjects learners study and the careers they may pursue in the future.

The system uses learners' Grade 9 subject marks, interests and career goals to provide a suggested academic stream. It uses Python and machine learning algorithms to analyse academic performance. It also includes a keyword-based Natural Language Processing (NLP) component that examines learners' written responses about their interests and career aspirations.

The project uses Decision Tree and Random Forest algorithms to analyse subject marks. The two models are compared using evaluation measures, and the better-performing model is selected according to the implemented model-selection process. The NLP component provides additional information based on keywords found in learners' responses.

A website developed using Flask connects the different components and allows learners to enter information and view their recommendations. The system is intended to support decision-making rather than make final educational decisions on behalf of learners.

This project demonstrates how Artificial Intelligence can be applied to an educational problem by combining data analysis, machine learning, text processing and web development in one application.

2. BUSINESS BACKGROUND AND RELEVANCE

2.1 Business Background

Learners completing Grade 9 must consider which subjects and academic streams they want to follow in Grade 10. Some learners find this decision difficult because they may not fully understand their strengths, interests or the relationship between school subjects and future careers.

Teachers, parents and school counsellors can provide guidance, but learners may need additional information to compare possible options. An online recommendation system can make this information easier to explore.

The proposed system uses information supplied by learners to produce a recommendation that can help them think about their academic choices. It combines subject performance with information about personal interests and career goals.

2.2 Relevance to the Project Theme

The theme of the project is **An AI Solution for Industries**. The education sector is the industry addressed by this project.

The system applies Artificial Intelligence to educational guidance by analysing learner information and providing personalised stream suggestions. It can help schools explore how technology may support learners, reduce the difficulty of comparing academic options and make educational information more accessible. The system does not replace teachers or professional career guidance. Instead, it provides an additional tool that learners and educators can use when discussing suitable study options.

2.3 Business Need

The main business need is to provide learners with an accessible way to explore possible Grade 10 academic streams using their academic performance, interests and career goals.

The solution also demonstrates how a school or educational organisation could use data-driven technology to support learner guidance. However, its recommendations must be evaluated carefully before it is used to influence real educational decisions.

3. PROBLEM DEFINITION

Grade 9 learners must make decisions about their Grade 10 academic subjects and possible future career paths. Some learners struggle to identify which academic stream best matches their school performance, interests and career goals. They may have limited access to personalised guidance or may not understand how their current subjects relate to future study and employment opportunities. This can make it difficult to compare available options and make informed decisions.

The AI-Based Learner Stream Recommendation System aims to address this problem by analysing learners' subject marks and written responses about their interests and career aspirations. The system uses machine learning to identify patterns in academic performance and keyword-based text processing to identify interests related to different academic streams. It then combines the available information to produce a suggested stream with an explanation.

The system focuses on four stream categories represented in the project dataset: Science, Commerce, Humanities and Technical/Vocational. These categories are used by the prototype and must be checked against the actual subject combinations offered by the learner's school.

The system is intended to support educational guidance, not to guarantee that a learner will succeed in a particular stream. Its usefulness depends on the quality of the dataset, the accuracy of the models and the reliability of the learner information supplied.

4. PROJECT AIM AND OBJECTIVES

4.1 Project Aim

The aim of the project is to develop an AI-based web application that helps Grade 9 learners explore suitable Grade 10 academic streams by analysing their subject marks, interests and career goals.

4.2 Project Objectives

The objectives are to:

1. Prepare and inspect a dataset containing learner subject marks and recommended stream labels.
2. Develop a Decision Tree model that predicts an academic stream from subject marks.
3. Develop a Random Forest model and compare its performance with the Decision Tree.
4. Evaluate the models using accuracy, precision, recall, F1-score and other appropriate evaluation methods.
5. Develop a keyword-based NLP component that analyses learners' interests and career goals.
6. Build a Flask website that collects learner information and displays recommendations.
7. Integrate the machine learning model, NLP component and website.
8. Test the system for functionality, usability and reliability.
9. Identify limitations, risks and possible improvements.
10. Use GitHub and GitHub Projects to manage project files and monitor progress.
11. Prepare the final documentation, presentation and project poster.

5. THEORETICAL ASPECTS

5.1 Artificial Intelligence

Artificial Intelligence (AI) refers to computer systems designed to perform tasks that normally require aspects of human intelligence. These tasks can include analysing information, identifying patterns, processing language and supporting decisions.

In this project, AI is used to analyse learner information and suggest an academic stream. The system uses machine learning to process subject marks and NLP techniques to analyse written interests and career goals.

AI is relevant because it allows the application to use information from learners to generate recommendations instead of requiring every learner to compare all possible options manually.

5.2 Machine Learning

Machine Learning is a branch of AI that allows computer systems to learn patterns from data and use those patterns to make predictions.

The project uses supervised machine learning. In supervised learning, a model is trained using examples that contain input information and known output labels. In this case, the input information consists of subject marks, while the output label represents an academic stream.

The model learns relationships between the subject marks and stream labels in the training dataset. It is then evaluated using separate test records to estimate how well it predicts examples that were not used during training.

The quality of the predictions depends on the amount, quality and representativeness of the training data.

5.3 Decision Tree Algorithm

A Decision Tree is a machine learning algorithm that makes predictions by following a series of decision rules. Each decision separates data according to selected features until the model reaches a predicted class.

For this project, the Decision Tree examines learners' subject marks and predicts one of the stream categories represented in the dataset.

One advantage of a Decision Tree is that its decision-making structure can be easier to explain than some more complex models. However, a tree that is too complex may learn patterns that do not generalise well to new learners.

The project uses a limited tree depth to help control complexity. Its performance must still be evaluated using appropriate test data.

5.4 Random Forest Algorithm

Random Forest is a machine learning algorithm that combines predictions from multiple decision trees. Each tree contributes to the final classification, helping the model capture different patterns in the data.

The project uses a Random Forest classifier to predict academic streams from learners' subject marks. The implemented configuration uses 200 trees and a maximum depth of six.

Random Forest can perform better than a single Decision Tree on some datasets, but this is not guaranteed. The two algorithms must be compared using the same data split and evaluation measures.

5.5 Natural Language Processing

Natural Language Processing, or NLP, is an area of AI that focuses on processing and analysing human language.

The project uses a rule-based, keyword-matching approach to analyse learners' written interests and career goals. For example, words related to technology, scientific investigation, business or creative activities may contribute to different stream scores, depending on the keyword rules implemented in the system.

This is a basic NLP approach. It does not fully understand the meaning of every sentence, and it may misunderstand spelling variations, unusual expressions or responses that contain several different interests.

The chatbot therefore provides additional information for the recommendation rather than a complete understanding of the learner.

5.6 Recommendation Systems

A recommendation system analyses available information to suggest options that may be relevant to a user.

In this project, the system combines information from the marks-based machine learning model and the NLP keyword component. When relevant keywords are found, the implemented recommendation logic uses a 60% machine learning and 40% NLP weighting. When relevant keywords are not found, the system falls back to the marks-based prediction.

The recommendation is a suggestion based on the information processed by the application. It is not a guarantee of academic success or a replacement for professional guidance.

5.7 Business Analysis

Business Analysis involves identifying a problem, understanding stakeholder needs, gathering requirements and determining how a proposed solution can address those needs.

In this project, the main stakeholders are Grade 9 learners, teachers, parents, school counsellors and educational institutions.

The analysis identifies the difficulty learners may experience when selecting academic streams and proposes a technology-based solution. The project requirements describe what the system should do, while testing helps determine whether the implemented features meet those requirements.

Business Analysis is important because a technically working application is not enough on its own. The solution must also address a real problem, be understandable to its users and provide useful information.

5.8 Project Management

Project Management involves planning, organising and monitoring activities to achieve a defined objective within the available time and resources.

The team divides the work into tasks covering project planning, data preparation, model development, NLP development, website development, integration, testing and documentation.

The project uses a work breakdown structure, task allocation, milestones and GitHub-based collaboration to monitor progress. These practices help the team identify delays, manage risks and keep track of changes.

5.9 Data Quality and Model Evaluation

Data quality refers to how suitable, accurate, complete and reliable information is for its intended purpose. A machine learning model cannot be expected to provide reliable predictions if its training data is inaccurate, unrepresentative or incorrectly labelled.

The project uses accuracy, precision, recall and F1-score to evaluate the classifiers. A confusion matrix can also help identify which stream categories are frequently confused.

Cross-validation can provide additional information about how model performance changes across different training subsets. However, cross-validation does not remove problems caused by poor data quality or data leakage.

Data leakage occurs when information that would not legitimately be available during prediction reveals the expected answer to a model. This is an important concern in the current project

because the dataset's chatbot responses may be related directly to the recommended stream labels.

5.10 Time Series Analysis

Time Series Analysis is a method of analysing data that is collected over time and arranged in time order. It is used to identify patterns such as trends, seasonal changes and movements between different time periods. For example, a school could use time series analysis to study how learner marks change from one term to another or how pass rates change over several years.

Time series methods normally require a time or date field together with repeated measurements of the same variable over different periods. The dataset used in this project contains 2,392 learner records with subject marks and a recommended stream label. The available project data is organised as learner records and does not provide a sequence of dated measurements for each learner. Therefore, the current system uses classification rather than a time series forecasting model.

Time Series Analysis was considered as part of the theoretical study, but it was not implemented in the current prototype because the available dataset does not contain the required time-based observations. It would not be correct to create a time series model from the existing records or to report forecasting results that were not produced. In a future version, if the school collects learner marks across several terms or years with reliable dates, time series analysis could be used to examine performance trends. Those trends could provide additional context for teachers and counsellors, while the final stream recommendation would still need to be validated using suitable educational data.

5.11 Deep Learning

Deep Learning is a branch of machine learning that uses artificial neural networks with multiple processing layers to learn patterns from data. These networks are inspired by the way connected neurons process information. Deep learning can be used for tasks such as image recognition, speech processing, language understanding and prediction when enough suitable training data and computing resources are available.

For a learner stream recommendation system, a possible deep learning approach is a feed-forward neural network. The network could receive the learner's subject marks as input, process them through one or more hidden layers and produce scores for the available stream categories. The category with the highest output score could then be considered as the model's predicted stream. The network would need to be trained and tested using correctly labelled learner data, with suitable measures for class imbalance and overfitting.

Deep learning is discussed as a possible approach, but it is not part of the current implementation. The project currently uses Decision Tree and Random Forest classifiers for marks-based prediction, together with a keyword-matching NLP component. These methods were selected for the prototype because they are suitable for structured tabular data and are easier to inspect with the available project setup. The current dataset also contains an imbalanced target, so a neural network would not automatically provide better results. It would need to be compared fairly with the existing models using the same training and test data, class-based precision and recall, F1-score, and cross-validation. Until such testing is completed, no deep learning accuracy or performance advantage can be claimed.

6. BUSINESS ANALYSIS AND REQUIREMENTS

6.1 Stakeholder Analysis

Grade 9 learners: The primary users who enter their subject marks and provide information about their interests and career goals.

Teachers: Can use the recommendations as an additional discussion point when helping learners consider their academic options.

Parents or guardians: May use the results to support discussions about academic strengths and future goals.

School counsellors: Can use the results as supplementary information during career and subject-choice guidance.

Educational institutions: May benefit from understanding how technology can support learner guidance, provided that the system is properly validated and used responsibly.

6.2 Functional Requirements

The system should:

1. Display a home page that explains the purpose of the application.
2. Allow learners to enter their Grade 9 subject marks.
3. Collect information about learners' interests and career goals.
4. Allow learners to select a school where the school-selection feature is available.
5. Process subject marks using the saved machine learning model.
6. Analyse written responses using the NLP keyword component.
7. Combine the available scores according to the implemented recommendation logic.
8. Display the suggested stream and an explanation of the result.
9. Show related career information and other stream scores where supported by the application.
10. Provide an About page explaining the system.
11. Handle invalid input and unexpected errors appropriately.

6.3 Non-Functional Requirements

Usability: The website should be easy to understand and navigate.

Performance: The system should process ordinary learner submissions within a reasonable time.

Reliability: The website should handle valid input consistently and provide useful feedback when errors occur.

Security: Sensitive information should be protected, and unnecessary personal information should not be collected.

Maintainability: The code should be organised so that the team can identify and update individual components.

Compatibility: The website should be tested on commonly used desktop and mobile screen sizes.

Explainability: Recommendations should include an understandable explanation and should not present a fit score as a guaranteed probability of success.

6.4 Hardware and Software Requirements

The development environment requires:

- A computer capable of running Python and a web browser.
- Python 3 and the required compatible libraries.
- Flask for the web application.
- scikit-learn for machine learning.
- Pandas for dataset handling.
- Joblib for saving and loading model files.
- A code editor such as Visual Studio Code.
- A web browser for testing the application.
- Git and GitHub for version control and collaboration.

The exact software versions should be recorded from the final development environment.

6.5 Business Rules

1. A learner must provide the subject marks required by the assessment form.
2. The application must validate input before processing it.
3. The recommendation must use the trained model and implemented NLP rules.
4. The system must explain that the result is a suggestion.

5. The available stream options must be checked against verified school information before practical educational decisions are made.
6. The application must not treat a high fit score as a guarantee of success.
7. Model performance must be reported honestly, including limitations and errors.

7. PROPOSED AI SOLUTION

7.1 Description of the Solution

The proposed solution is a web-based recommendation system that combines machine learning, keyword-based text processing and a user-friendly interface.

The learner enters their Grade 9 marks and provides information about interests and career goals. The application processes the information and generates a suggested academic stream.

The machine learning component uses the marks to predict a stream. The NLP component identifies keywords in the learner's responses and calculates stream scores. The recommendation component combines these results according to the implemented rules and sends the result to the website for display.

7.2 How the System Works

Step 1: Learner opens the website.
The home page introduces the purpose of the application and explains how the recommendation process works.

Step 2: Learner provides academic information.
The learner enters the required Grade 9 subject marks.

Step 3: Learner provides interests and career goals.
The learner responds to questions about favourite subjects, interests and possible future careers.

Step 4: The machine learning model processes the marks.
The saved model analyses the subject marks and produces a stream prediction.

Step 5: The NLP component analyses the responses.
The keyword rules search for words associated with the stream categories and calculate additional scores.

Step 6: The recommendation component combines the results.
When relevant keywords are identified, the system applies its configured weighting. If no relevant keywords are identified, it uses the marks-based prediction.

Step 7: The website displays the recommendation.
The result page presents the suggested stream, fit score, explanation and related career information where available.

Step 8: The learner considers the available options.
The learner can use the result as a starting point for discussion with teachers, parents or a school counsellor.

7.3 System Flow

Learner input → Data validation → Marks-based prediction → NLP keyword analysis → Recommendation combination → Results page.

The system flow describes the intended process. The accuracy and usefulness of the final recommendation depend on proper testing and validation.

8. PROJECT SCOPE

8.1 Included in the Project

The project includes:

- A web interface developed using Flask.
- A dataset containing learner performance information.
- Decision Tree and Random Forest classifiers.
- Model evaluation and comparison.
- A keyword-based NLP chatbot component.
- A recommendation module that combines model and text scores.
- An optional school-selection feature.
- A recommendation results page.
- Functional testing and documentation.
- GitHub-based project collaboration.

8.2 Outside the Current Scope

The current prototype does not establish that its recommendations are suitable for every learner or every South African school.

It does not replace professional career counselling, school admission decisions or official subject-selection requirements. It also does not demonstrate that the model has been validated using a sufficiently large, independently collected sample of real learners.

The current implementation should therefore be treated as a prototype that requires further evaluation before real-world deployment.

9. PROJECT TEAM STRUCTURE

9.1 Roles and Responsibilities

Comfort — Project Manager

- Coordinate the project activities.
- Allocate tasks to team members.
- Monitor deadlines and progress.
- Organise communication and meetings.
- Monitor project risks.
- Coordinate integration and final delivery.

Tebogo Mathiba — Data and Dataset Developer

- Inspect the dataset.
- Check missing values, duplicates and data quality.
- Prepare the data for model development.
- Check the training and testing split.
- Record the dataset's source and limitations.

Botshelo Sebati — Machine Learning Developer

- Develop the Decision Tree and Random Forest classifiers.
- Train and compare the models.
- Evaluate the model results.
- Save the selected model and label encoder.
- Assist with investigating poor performance and possible data leakage.

Rethabile Ramodise — NLP and Chatbot Developer

- Develop the keyword-matching rules.
- Process learner interests and career goals.
- Test the chatbot using different responses.
- Review how keyword matches affect recommendations.
- Assist with integration of NLP scores.

Hitekani Mdaka — Web Developer

- Develop the Flask website pages and forms.
- Connect the website to the recommendation components.
- Display recommendations and explanations.
- Check navigation and input handling.
- Support the integration of the system components.

Ndumiso Duma — Testing, Documentation and UI/UX

- Prepare test cases.
- Test the website and recommendation process.
- Record errors and assist with improvements.
- Review usability and interface consistency.
- Assist with documentation and presentation preparation.

The assigned roles describe the planned division of work. The team should update the record to reflect the actual contributions completed by each member.

10. WORK BREAKDOWN STRUCTURE

The project is divided into the following work packages.

10.1 Project Management

- Define the problem and objectives.
- Identify project requirements.
- Allocate responsibilities.
- Monitor progress.

- Manage risks and deadlines.

10.2 Data Preparation

- Inspect the CSV dataset.
- Review the data fields and stream labels.
- Check missing values and duplicates.
- Examine mark ranges and class distribution.
- Confirm the training and testing split.
- Document data limitations.

10.3 Machine Learning Development

- Prepare the input features and target labels.
- Train the Decision Tree model.
- Train the Random Forest model.
- Evaluate the classifiers.
- Compare their results.
- Save the selected model and label encoder.

10.4 NLP and Chatbot Development

- Identify relevant interest and career keywords.
- Create keyword-matching rules.
- Calculate stream scores.
- Test responses with different wording.
- Test responses without matching keywords.
- Investigate possible data leakage.

10.5 Website Development

- Develop the Flask routes and templates.
- Create the learner input forms.
- Connect the saved model.
- Connect the NLP and recommendation modules.

- Display results and explanations.
- Test page navigation and input validation.

10.6 Integration and Testing

- Connect all system components.
- Test the complete learner journey.
- Check for errors during submission.
- Evaluate model performance.
- Record issues and corrections.
- Document the final limitations.

10.7 Documentation and Presentation

- Prepare the project report.
- Verify the GitHub repository.
- Prepare the presentation and poster.
- Complete the originality declaration.
- Attach the required Grammarly evidence.
- Prepare the final PDF for submission.

11. PROJECT SCHEDULE

The project plan uses a five-week development structure. The dates below should be updated to match the team's actual work and evidence.

Week 1: Requirements and Dataset Review

Activities include defining the problem, identifying stakeholders, gathering requirements, allocating tasks and inspecting the dataset.

Expected milestone: A documented project plan and initial dataset review.

Week 2: Machine Learning and NLP

Activities include training the classifiers, comparing initial results and developing the keyword-based chatbot.

Expected milestone: Initial machine learning and NLP components.

Week 3: Website Development

Activities include creating Flask pages, learner forms, navigation and the recommendation interface.

Expected milestone: Working website pages.

Week 4: Integration

Activities include connecting the model, chatbot, recommendation module and website.

Expected milestone: An integrated application that can process learner information.

Week 5: Testing and Finalisation

Activities include functional testing, model evaluation, error correction, documentation, poster preparation and presentation preparation.

Expected milestone: A tested prototype and final project submission materials.

Progress should be tracked using GitHub Projects. Actual completion dates and evidence should be recorded rather than assuming every milestone has been completed.

12. TOOLS AND TECHNOLOGIES

12.1 Python

Python is the main programming language used for the recommendation logic, data processing and Flask backend. It provides libraries that support machine learning and web application development.

12.2 scikit-learn

Scikit-learn provides the machine learning algorithms used in the project. It supports Decision Tree and Random Forest classifiers, model evaluation and cross-validation.

12.3 Pandas

Pandas is used to read and inspect the CSV dataset and prepare the information required by the machine learning code.

12.4 Joblib

Joblib is used to save and load the trained model and label encoder so that the website can reuse the trained components.

12.5 Flask

Flask provides the web application backend. It handles website routes, receives user input and connects the interface to the recommendation logic.

12.6 HTML and CSS

HTML provides the structure of the web pages, while CSS controls the visual design, layout and presentation.

12.7 JavaScript

JavaScript supports client-side interactions where implemented, such as interactive website behaviour.

12.8 Git and GitHub

Git tracks changes to the project files. GitHub provides an online repository for collaboration, version control and project evidence.

The project repository is:

https://github.com/Comfort-RSA/BA-PROJECT-2026

The team should verify that the repository contains the final code and that its GitHub Projects board accurately reflects the work completed.

12.9 Main Project Files

The supplied implementation includes the following important files and components:

- Student_performance_data_.csv — learner dataset.
- stream_model.pkl — saved machine learning model.
- label_encoder.pkl — saved label encoder.
- nlp_chatbot.py — keyword-based text analysis.
- combine_results.py — combines machine learning and NLP scores.
- schools.json — school information.
- app.py — Flask application entry point.

The files should be checked against the final repository before submission.

13. SYSTEM DESIGN AND IMPLEMENTATION

13.1 Dataset

The supplied dataset is named Student_performance_data_.csv. The current project documentation records 2,392 rows, with 1,435 training records and 957 test records.

The input features include nine subject-mark fields:

- Mathematics
- Life Orientation
- First Additional Language
- Social Science
- EMS
- Home Language
- Creative Arts
- Natural Science
- Technology

The target column is Recommended_Stream, which represents the stream label that the model attempts to predict.

The dataset also contains fields associated with student identification, the data split and chatbot responses. These fields must be handled carefully so that the target label is not accidentally revealed to the model.

The source, collection method, representativeness and labelling process of the dataset must be documented. If the records are synthetic or generated for demonstration, the report must state this clearly.

13.2 Machine Learning Implementation

The implementation trains and compares a Decision Tree classifier and a Random Forest classifier.

The Decision Tree configuration uses a maximum depth of five. The Random Forest configuration uses 200 trees and a maximum depth of six. Both use a fixed random state in the recorded configuration to support repeatable experiments.

The model-selection process compares test accuracy and uses F1-score as a tie-breaker. The selected model is saved for use by the website.

These settings describe the supplied implementation. They do not, by themselves, prove that the model is accurate or suitable for real-world use.

13.3 NLP Implementation

The NLP component searches the learner's written responses for keywords related to the available stream categories.

The component produces stream scores based on the matched keywords. These scores are passed to the recommendation module.

This approach is relatively simple to implement and explain, but it has limitations. A learner may describe an interest using words that are not included in the keyword lists, and a word may have different meanings depending on its context.

The component therefore requires testing with different responses, spelling variations and ambiguous statements.

13.4 Recommendation Integration

The recommendation component combines the marks-based machine learning output and NLP scores.

According to the supplied implementation, the machine learning component contributes 60% and the NLP component contributes 40% when relevant keywords are found. If no relevant keywords are found, the system falls back to the marks-based prediction.

The recommendation is then passed to the Flask website, which displays the suggested stream and supporting information.

13.5 Website Implementation

The website provides the interface through which learners interact with the system.

The main pages include:

- Home page.
- Optional school-selection page.
- Subject-mark assessment page.
- Chatbot or interest-question page.
- Recommendation results page.
- About page.

The website connects the learner input forms to the backend recommendation process. The displayed result can include the suggested stream, fit score, explanation, related careers and other stream scores.

School information should be verified before it is presented as official guidance. The application should also be reviewed for secure configuration before public deployment.

14. TESTING AND EVALUATION

14.1 Purpose of Testing

Testing checks whether the website works as intended and whether the recommendation process produces results that can be evaluated.

The project requires both functional testing and model evaluation. Functional testing checks website behaviour, while model evaluation measures prediction performance.

14.2 Initial Model Results

The following results were recorded in the supplied project's console output. They describe the current implementation and should not be treated as final proof of real-world accuracy.

Evaluation item	Recorded result
Decision Tree test accuracy	41.17%
Decision Tree weighted precision	43.33%
Decision Tree weighted recall	41.17%
Decision Tree weighted F1-score	41.31%
Random Forest test accuracy	41.38%
Random Forest weighted precision	42.98%
Random Forest weighted recall	41.38%
Random Forest weighted F1-score	41.46%
Random Forest five-fold cross-validation accuracy	31.57% mean
Cross-validation standard deviation	9.42 percentage points
NLP-only validation	100% on 957 test rows
Combined validation	99.9% on 957 test rows

The Decision Tree and Random Forest results indicate that the marks-only models have limited predictive performance on the recorded test split. The Random Forest performed slightly better in this run, but the difference is small.

The cross-validation result is lower than the held-out test accuracy. This difference should be investigated because it may indicate instability in model performance or differences between the training and testing data.

14.3 Important Data Leakage Concern

The NLP-only validation recorded 100%, while the combined validation recorded 99.9%. These high figures require careful investigation.

The supplied validation approach uses chatbot responses from dataset records whose stream labels are also being predicted. If the responses were generated using the target stream labels, or contain words that reveal those labels, the NLP component may effectively be receiving information about the correct answer.

This is known as data leakage. It can make a model appear more accurate than it would be when processing responses from new learners.

Therefore, the NLP-only and combined results must not be presented as proof of reliable real-world performance until the leakage risk has been investigated and the evaluation has been repeated using suitable independent data.

14.4 Evaluation Measures

Accuracy: Measures the proportion of predictions that are correct.

Precision: Measures how often predictions for a particular class are correct.

Recall: Measures how many actual examples of a class are correctly identified.

F1-score: Combines precision and recall into one measure.

Confusion matrix: Shows the correct and incorrect predictions for each class.

Cross-validation: Repeats training and evaluation across different data folds to help examine the consistency of model performance.

Using several evaluation measures gives a clearer picture than relying on accuracy alone.

14.5 Functional Test Plan

Test case	Expected result

Open the home page	The page loads and explains the system
Enter valid subject marks	The form accepts the values
Submit invalid marks	The system provides appropriate validation feedback
Submit learner interests	The NLP component processes the text
Submit responses with no matching keywords	The marks-based fallback is used
Generate a recommendation	The result page displays the suggested stream
Load the saved model	The application processes a prediction without a model-loading error
Navigate between pages	The website routes work correctly
Select a school where supported	The application displays the available recorded school information
Test the website on different screen sizes	The interface remains usable

This is a test plan. The team should record the actual outcome of every test as **Pass**, **Fail** or **Not Tested**, together with evidence. Expected results must not be described as completed tests unless they have been performed.

14.6 Recommended Improvements

The next stage of evaluation should include:

1. Confirming how the dataset and chatbot responses were created.
2. Checking whether any input field reveals the target stream.
3. Ensuring that training and testing data are separated correctly.
4. Testing the system using independently collected learner responses where appropriate and ethically approved.
5. Comparing marks-only and combined predictions on the same valid test set.
6. Reviewing performance separately for each stream category.
7. Testing the application with different input combinations.

8. Recording errors and correcting them before final deployment.

15. PROJECT RISK MANAGEMENT

Risk	Impact	Mitigation
Insufficient or unrepresentative data	High	Document the data source and limitations, and seek appropriate verified data where possible
Low machine learning accuracy	High	Review data quality, labels, features and model configurations
Data leakage	High	Check how stream labels and chatbot responses were created and evaluate with independent data
Incorrect NLP keyword matches	Medium	Test different wording, spelling variations and ambiguous responses
Website and model integration errors	High	Test individual components and the complete learner journey
Incorrect school information	Medium	Verify school information and indicate when details are unavailable
Project delays	High	Monitor tasks and deadlines using GitHub Projects
Security or privacy problems	High	Collect only necessary information and protect application secrets
Poor usability	Medium	Test the interface and improve navigation and explanations
Incomplete submission evidence	High	Check the documentation, signed declaration, poster, presentation, Grammarly evidence and final PDF

Risk management is important because the project combines data, software, user input and educational guidance. The team should review these risks regularly and record any corrective actions.

16. PROJECT CONSTRAINTS AND LIMITATIONS

16.1 Dataset Limitations

The quality of the recommendation depends on the dataset. If the data does not represent a broad range of learners, schools or academic backgrounds, the predictions may not generalise to all learners.

The source and nature of the supplied dataset must be verified and described accurately.

16.2 Model Performance

The recorded marks-only accuracy is approximately 41%. This indicates that the current models require improvement before their predictions can be considered dependable for practical educational guidance.

The project should report this result honestly rather than presenting the prototype as a highly accurate system.

16.3 NLP Limitations

The keyword-based NLP component uses predefined rules. It may not understand context, spelling errors, synonyms or complex statements. It can also produce misleading scores when a learner uses words associated with several streams.

16.4 School Differences

Schools may offer different subjects, combinations and academic pathways. A stream category in the dataset may not match the options available at every school.

The system should therefore verify available subject combinations before presenting a recommendation as a practical option for a particular learner.

16.5 Time and Resources

The project must be completed within the allocated academic period. The team has limited time to collect data, develop the application, test the models, improve the interface and prepare the required documentation.

16.6 Privacy and Ethical Considerations

Learner information should be handled responsibly. The team should minimise the collection of personal information, protect any stored data and avoid exposing identifiable learner records in the public repository.

If real learner information is collected for further testing, the team must follow the relevant consent, institutional approval and privacy requirements.

16.7 Prototype Status

The current application demonstrates the main idea and integrates the website, machine learning and NLP components. However, the model evaluation and possible data leakage concerns mean that additional work is required before the system can be recommended for real-world educational decisions.

17. EXPECTED BENEFITS AND SUCCESS CRITERIA

17.1 Expected Benefits

Improved access to guidance: Learners can explore stream options through a website.

Personalised suggestions: The system uses individual marks and responses to generate recommendations.

Better awareness: Learners can reflect on the relationship between academic strengths, interests and future careers.

Support for educators: The recommendation can serve as an additional discussion point during subject-choice guidance.

Practical use of AI: The project demonstrates how machine learning and NLP can be combined in an educational application.

Opportunities for improvement: The prototype provides a starting point for further research into educational recommendation systems.

These are intended benefits. Their actual impact on learner decision-making must be evaluated before it can be claimed as a demonstrated outcome.

17.2 Success Criteria

The project will meet its functional objectives when:

1. Learners can access the website and navigate its pages.
2. Valid subject marks can be submitted successfully.
3. The application can load the saved model and generate a prediction.
4. The chatbot can process written interests and career goals.
5. The recommendation module can combine the available outputs.
6. The results page can display the recommendation and explanation.
7. Functional tests are completed and documented.
8. Model evaluation is conducted using an appropriate testing approach.

9. Data leakage concerns are investigated and reported.
10. The GitHub repository contains the correct project files.
11. The final documentation, presentation, poster and required supporting evidence are prepared.

18. PROJECT DELIVERABLES

The project deliverables are:

1. Project planning and requirements documentation.
2. Dataset description and data-quality review.
3. Machine learning implementation using Decision Tree and Random Forest.
4. Saved model and label encoder.
5. NLP keyword-matching chatbot.
6. Recommendation combination module.
7. Flask website and templates.
8. Functional test plan and recorded results.
9. Model evaluation report and discussion of limitations.
10. GitHub repository and GitHub Projects evidence.
11. Final project documentation.
12. Project presentation.
13. Soft-copy project poster.
14. Signed declaration of originality from all group members.
15. Required Grammarly report or certificate evidence.

Any deployment link should be included only if the application has actually been deployed and tested. The repository and supporting evidence should be checked before the final submission.

19. CONCLUSION

The AI-Based Learner Stream Recommendation System demonstrates how Artificial Intelligence can be used to support educational decision-making. The project aims to help Grade 9 learners explore possible Grade 10 academic streams by considering their subject marks, interests and career goals. The prototype combines a Flask website, Decision Tree and

Random Forest classifiers, a keyword-based NLP chatbot and a recommendation component to provide learners with suggested academic streams and supporting information.

The initial results show that the marks-only models achieved approximately 41% accuracy on the recorded test data. Although the NLP-only and combined validation results were much higher, they may have been affected by data leakage because of the relationship between chatbot responses and stream labels. These results need further investigation before the system's real-world accuracy can be confirmed. The project also highlights the importance of data quality, proper testing, risk management and responsible AI use.

In conclusion, the prototype provides a foundation for developing an AI-based educational guidance system. Future improvements should focus on using better-validated data, conducting independent testing, improving model performance, enhancing language processing and verifying school subject information. The system should support learners in making informed decisions, while final subject choices should be made with guidance from teachers, parents and school counsellors.

20. REFERENCES

1. Pedregosa, F. et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, pp. 2825–2830. Available at: https://jmlr.org/papers/v12/pedregosa11a.html
2. Pallets Projects. (n.d.). *Flask Documentation*. Available at: https://flask.palletsprojects.com/
3. Python Software Foundation. (n.d.). *Python 3 Documentation*. Available at: https://docs.python.org/3/
4. The pandas development team. (n.d.). *pandas Documentation*. Available at: https://pandas.pydata.org/docs/
5. Comfort-RSA. (2026). *BA-PROJECT-2026: AI-Based Learner Stream Recommendation System*. GitHub repository. Available at: https://github.com/Comfort-RSA/BA-PROJECT-2026

Hyndman, R.J. and Athanasopoulos, G. (2021). Forecasting: Principles and Practice. 3rd ed. Available at: https://otexts.com/fpp3/

Goodfellow, I., Bengio, Y. and Courville, A. (2016). Deep Learning. Cambridge, MA: MIT Press. Available at: https://www.deeplearningbook.org/